120 Days of Quantitative Finance

Month 4: Stochastic Calculus

Week 14: SDEs and Black-Scholes

Day 98: Black-Scholes Implementation

1 Introduction

The Black-Scholes model provides a closed-form solution for pricing European options.

Alternatively, Monte Carlo simulation offers a numerical approach based on simulated price paths.

2 Closed-Form Solution

The Black-Scholes formula allows direct computation of option prices under model assumptions.

Advantages:

- fast computation
- exact under model assumptions

Limitations:

- restricted to simple payoffs
- assumes constant volatility

3 Monte Carlo Simulation

Monte Carlo estimates option prices by:

1. Simulating multiple asset price paths
2. Computing payoff at maturity
3. Averaging and discounting

4 Comparison

- Closed-form: efficient and precise
- Monte Carlo: flexible and general

5 Trade-Offs

- Accuracy vs computational cost
- Simplicity vs flexibility

6 Financial Insight

Monte Carlo methods are essential when:

- payoffs are path-dependent
- models are complex
- no analytical solution exists

7 Conclusion

Both approaches are fundamental tools in quantitative finance.

Understanding when to use each is critical for practical modeling.